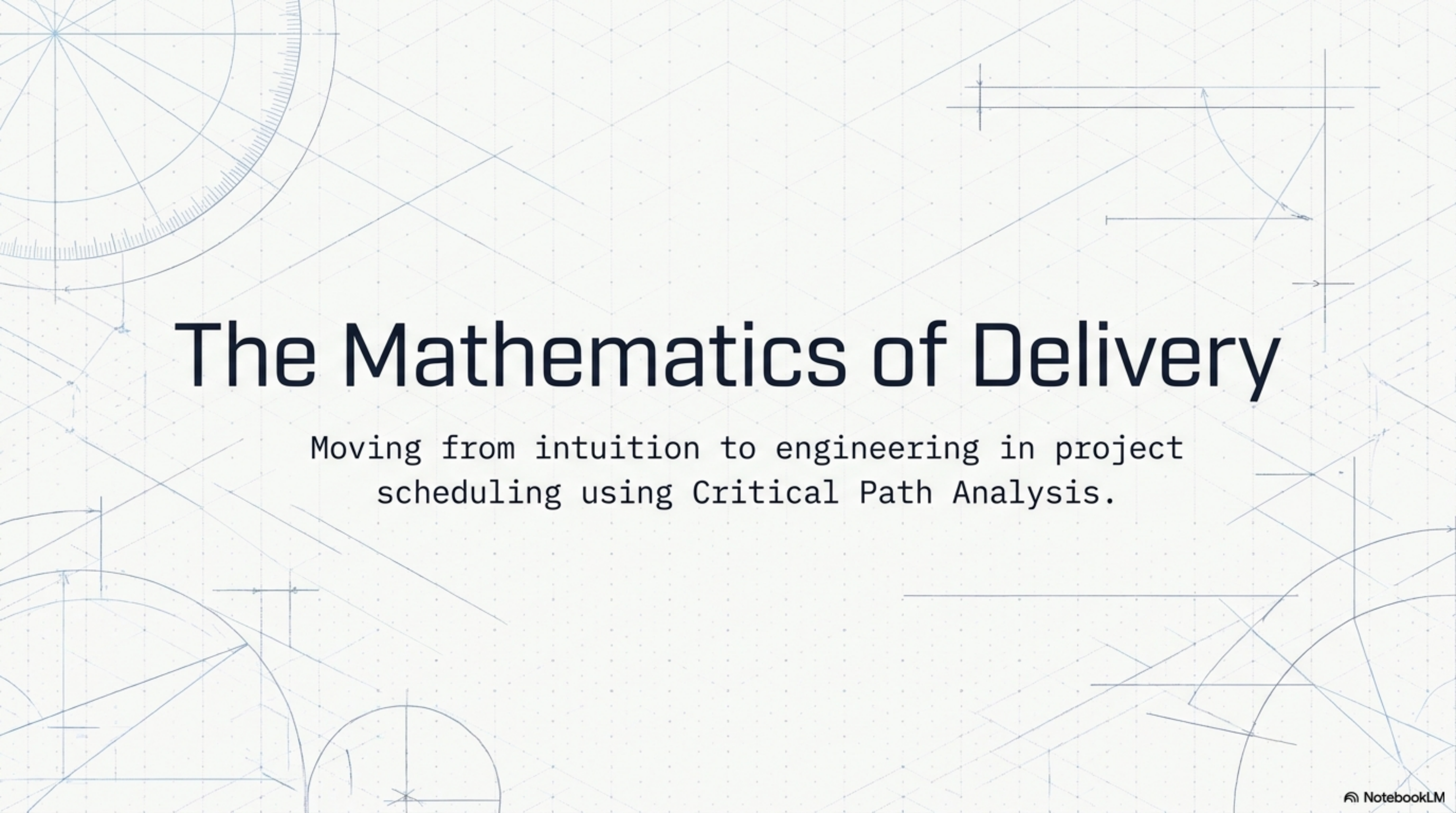
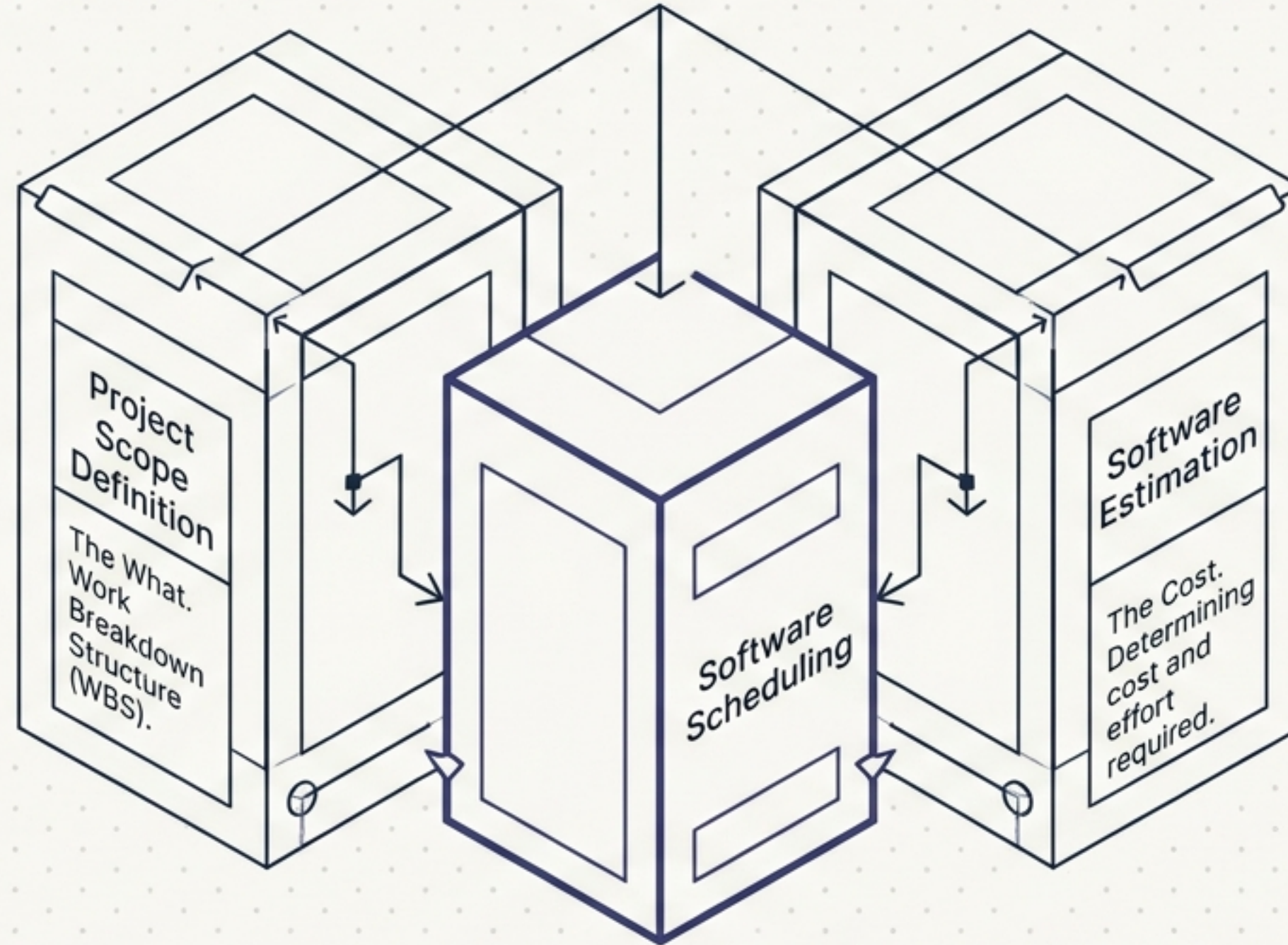


The background features a light blue grid with various geometric drawings in a darker blue. These include a circular sector with radial lines in the top-left, a coordinate system with a curve in the top-right, and several intersecting lines and arcs in the bottom-left and bottom-right.

The Mathematics of Delivery

Moving from intuition to engineering in project scheduling using Critical Path Analysis.

Project planning is a triad of scope, effort, and time.



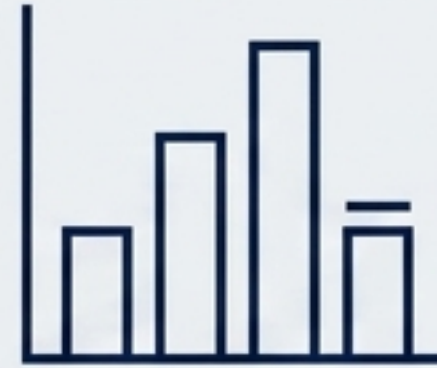
The Time. Organizing tasks concurrently to optimize workforce and minimize dependencies.

The Four Laws of Project Delay



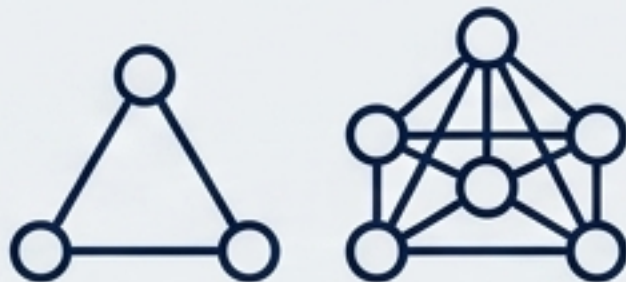
The Estimation Trap

Predicting the exact difficulty and cost of solving novel problems is inherently flawed.



The Productivity Myth

Output is never strictly proportional to the number of people assigned to a task.



3 Nodes, 3 Lines

4 Nodes, 6 Lines

Brooks's Law

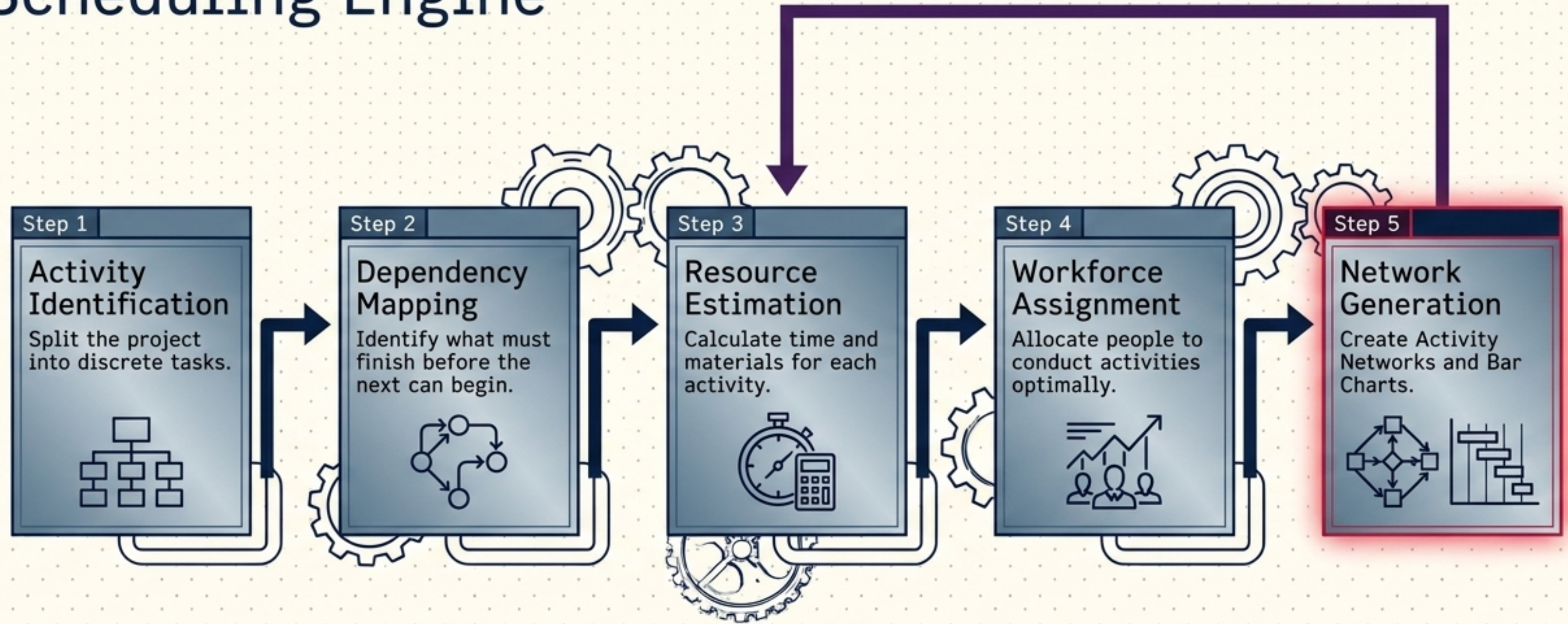
Adding people to a late project makes it later due to exponential communication overhead.



The Contingency Mandate

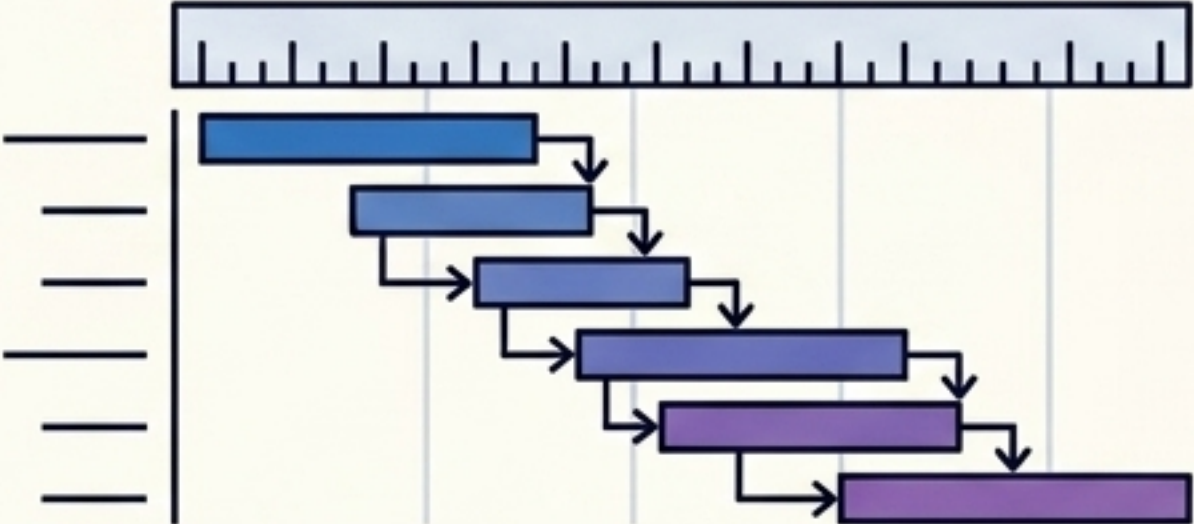
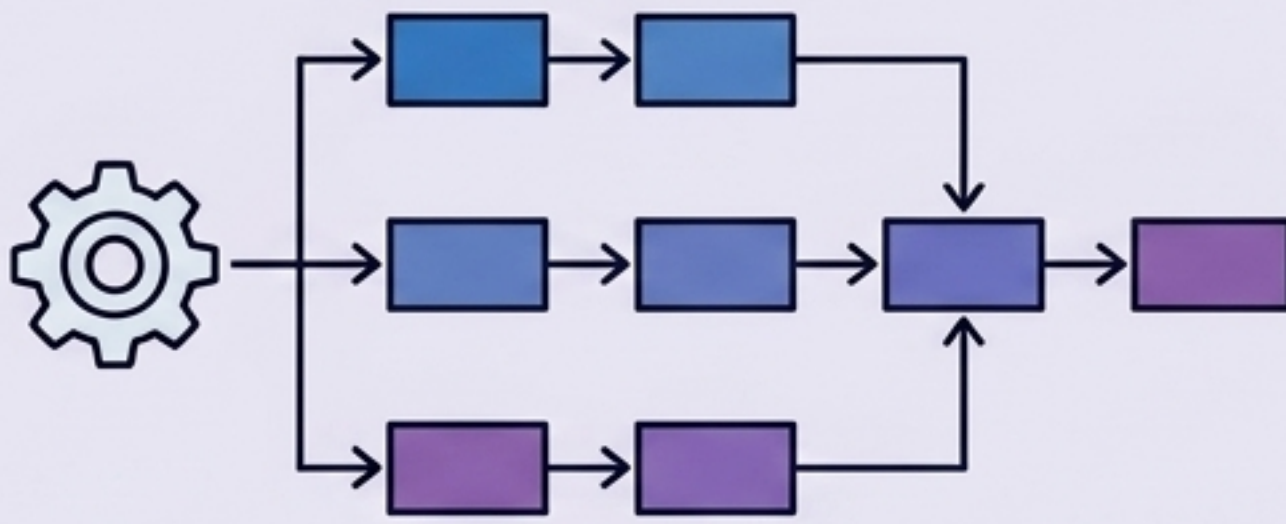
The unexpected is guaranteed. Planning without contingency is planning for failure.

The Project Scheduling Engine



Selecting the Right Tool:

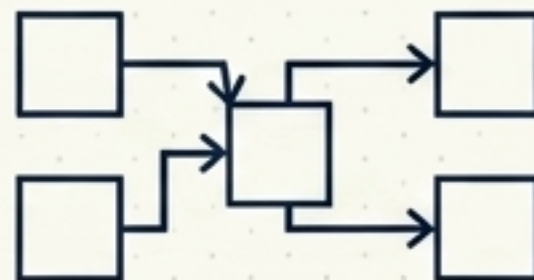
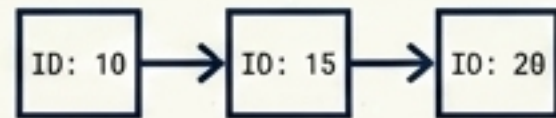
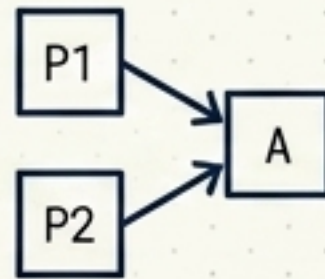
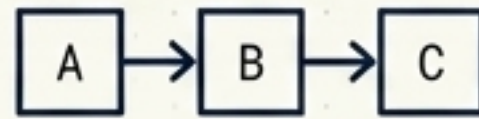
Bar Charts vs. Activity Networks

Bar (Gantt) Charts		Activity Networks (CPM/PERT)	
			
Primary Focus: ID: 7	Calendar Time.	Primary Focus: ID: 9, 10, 22	Sequential Dependencies.
Visual Layout: ID: 1	Horizontal Timelines.	Visual Layout: ID: 10, 11, 24	Activity-on-Arrow/Node webs.
Best Use Case: ID: 3	High-level stakeholder tracking.	Best Use Case: ID: 16, 27	Complex bottleneck planning and mathematical analysis.
Key Strength: ID: 3	Highly intuitive to read schedule against a calendar.	Key Strength: ID: 24, 23	Mathematical rigor in identifying critical delays.

The Architecture of Activity Networks

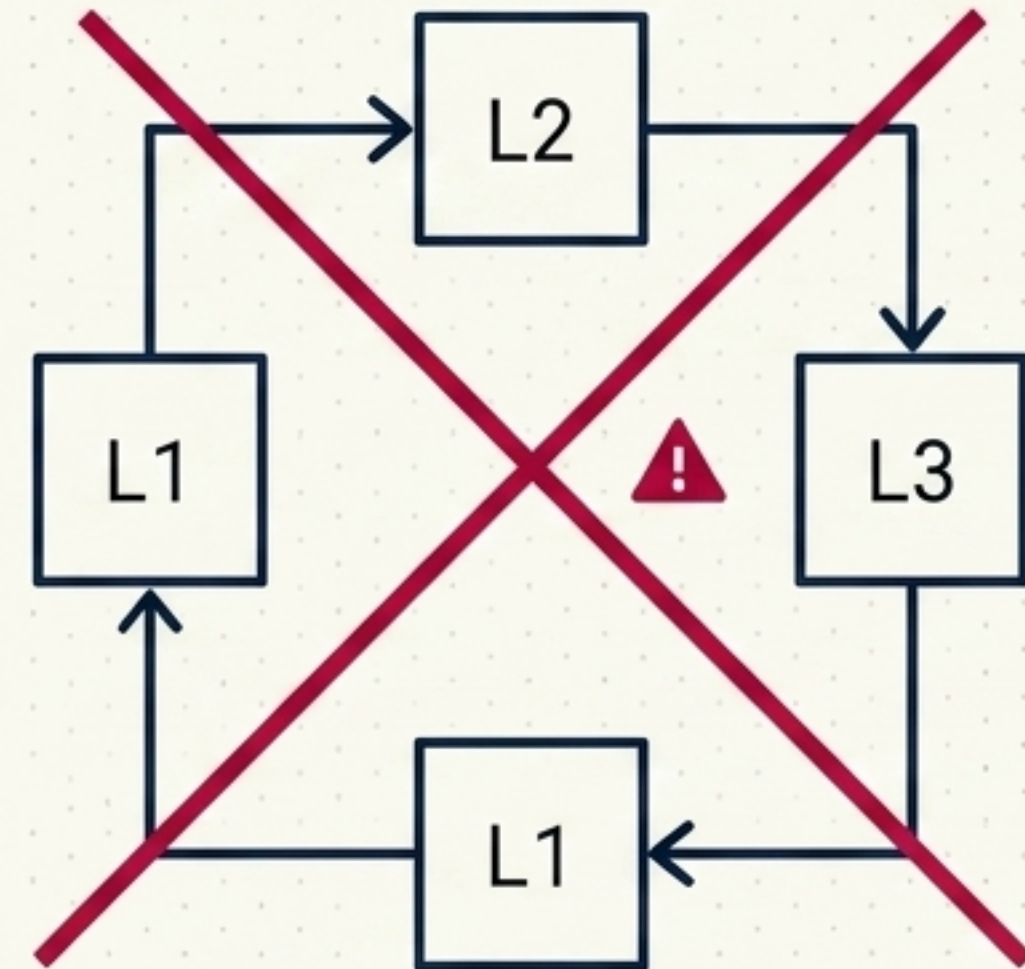
Structural Rules (Do's)

- ✓ **Direction**
Flow must always be from left to right.
- ✓ **Prerequisites**
An activity cannot begin until ALL predecessors are completed.
- ✓ **Identification**
Every activity needs a unique ID, and an ID must be mathematically greater than its preceding activity.
- ✓ **Geometry**
Arrows can overlap to show project flow.

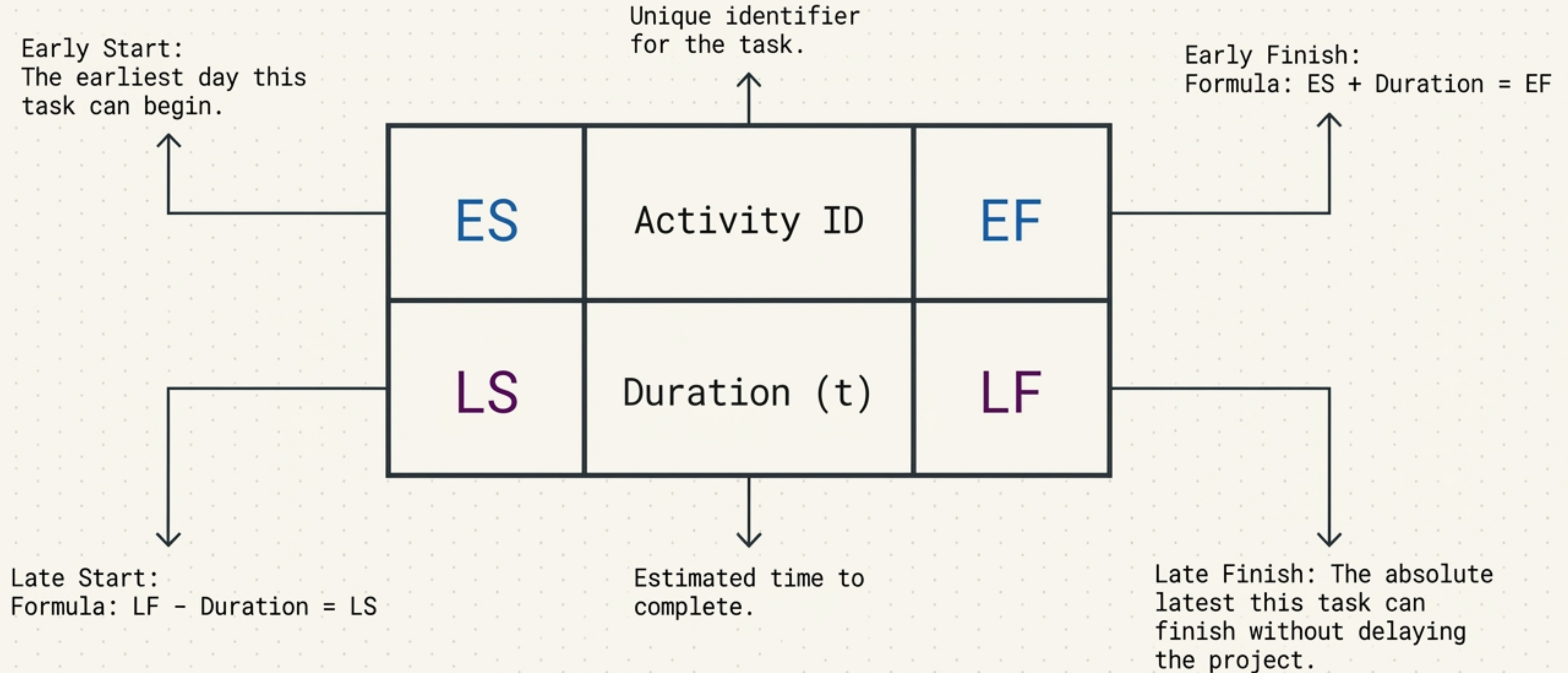


Strict Constraints (Don'ts)

- ✗ **Constraint**
Looping is strictly prohibited.

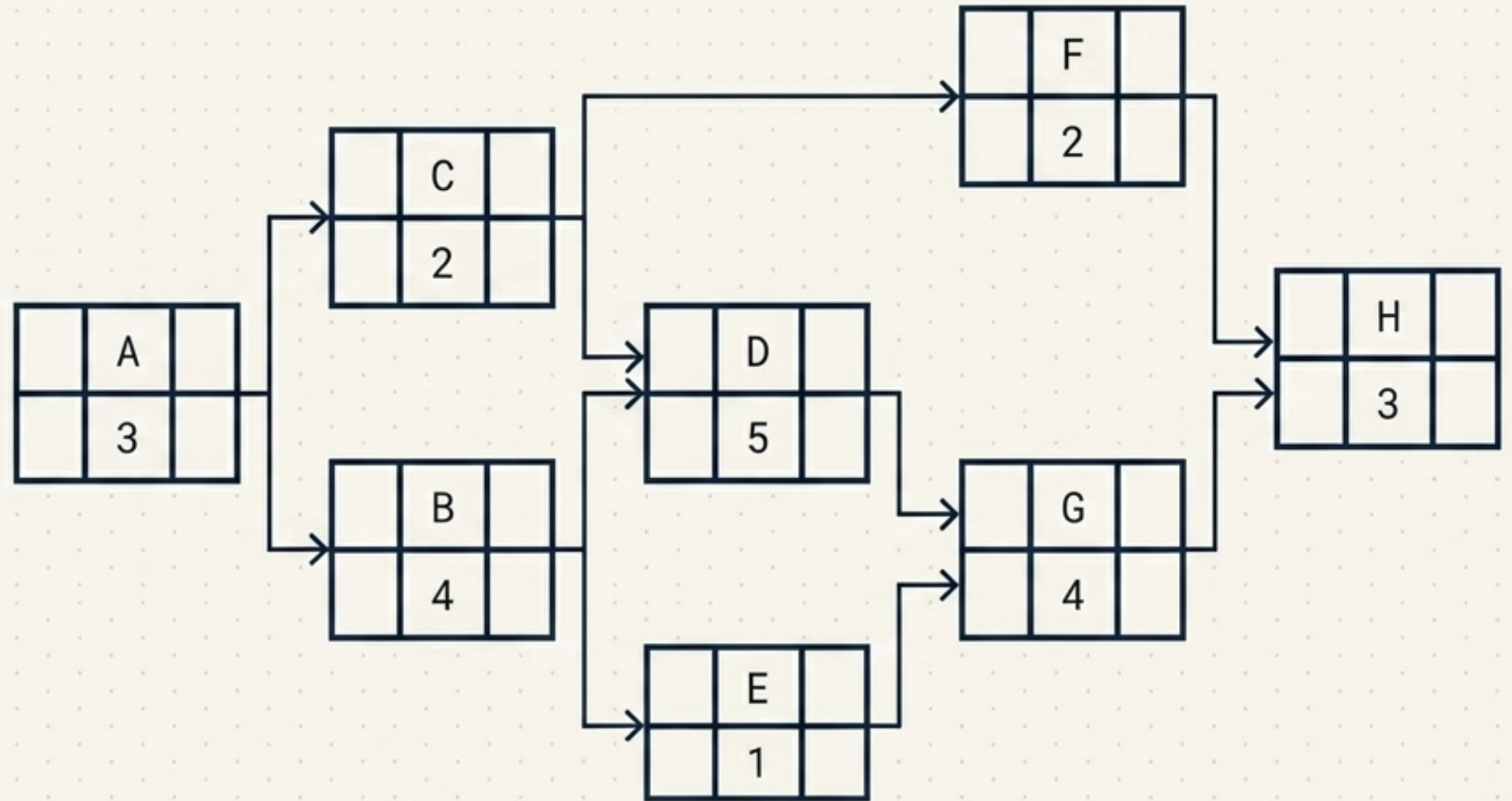


Anatomy of a Project Node

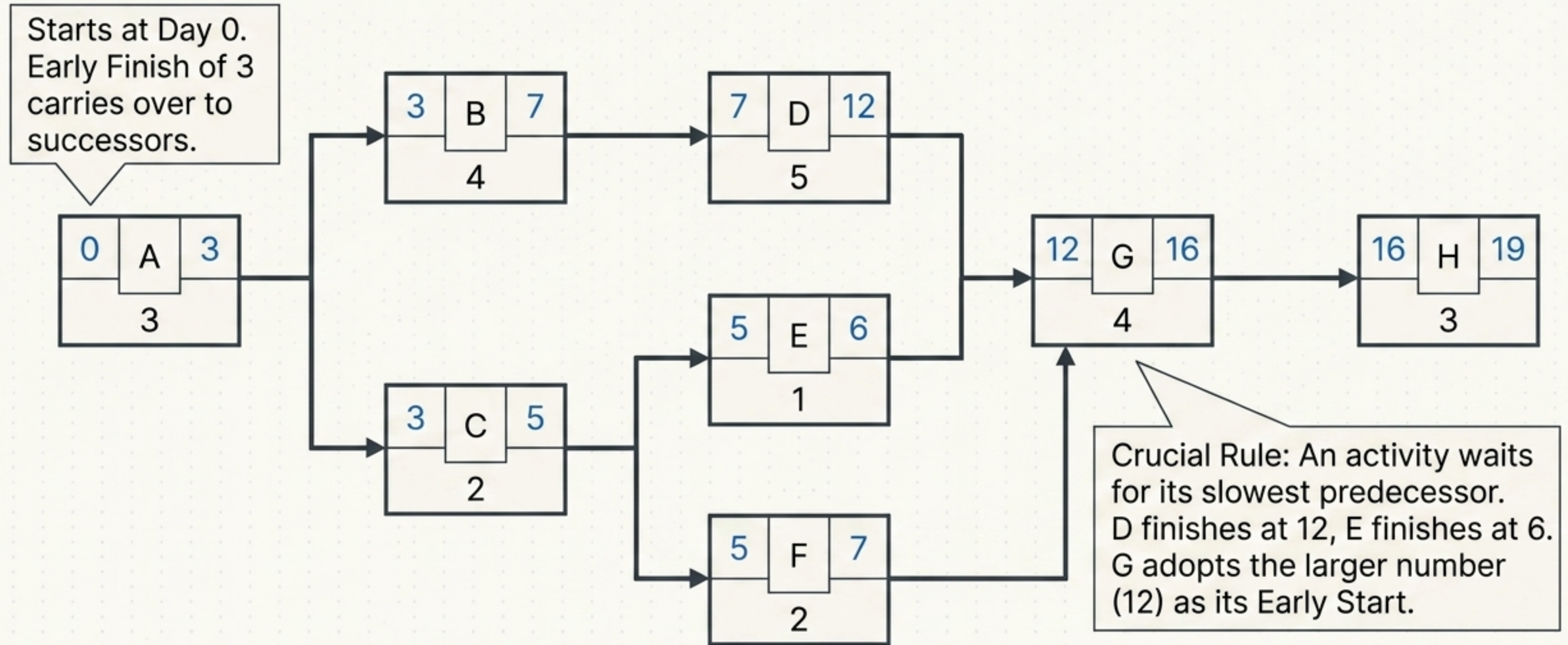


Translating Data into Structure

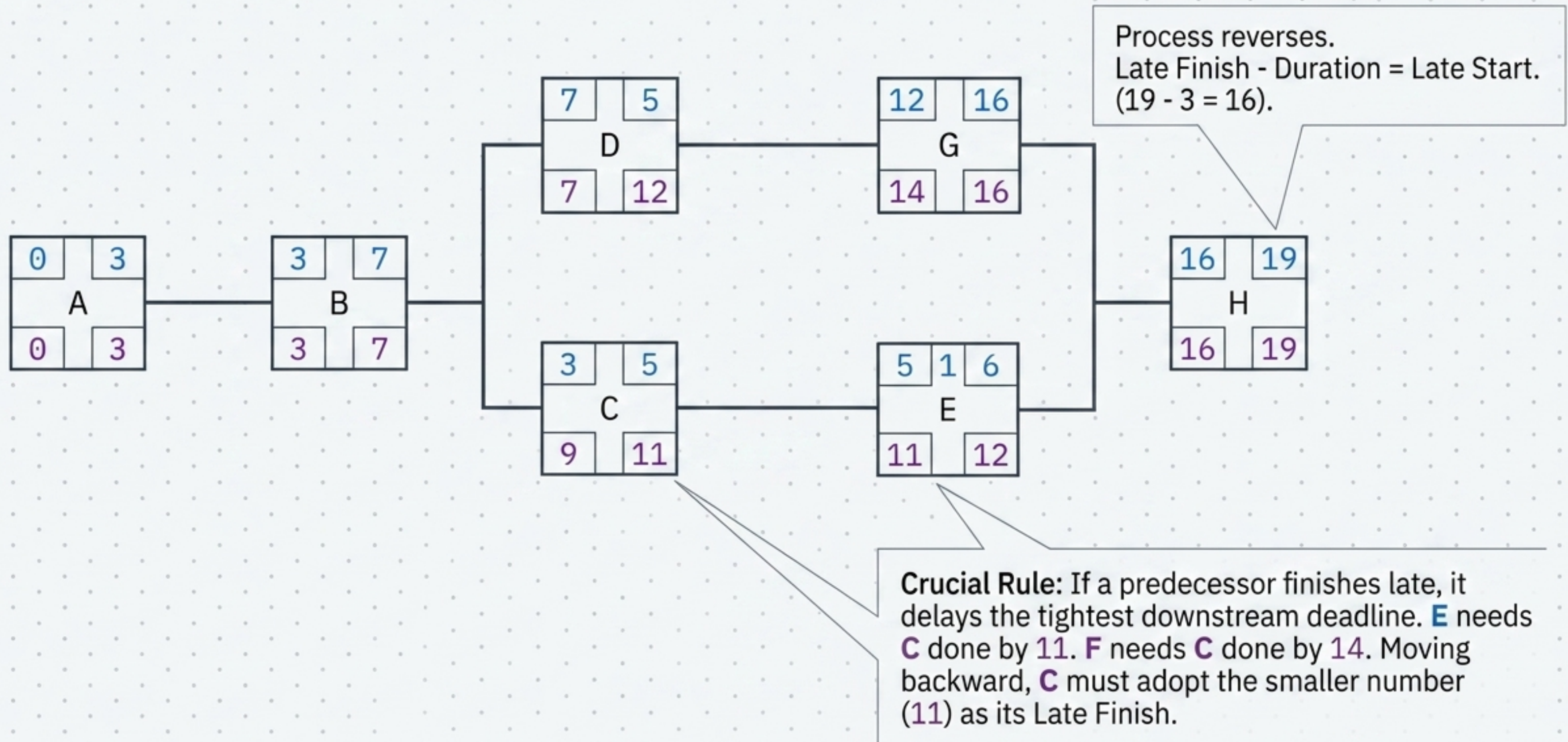
Activity	Predecessor	Duration
		(days)
A	-	3
B	A	4
C	A	2
D	B	5
E	C	1
E	C	2
F	C	2
G	D, E	4
H	F, G	3



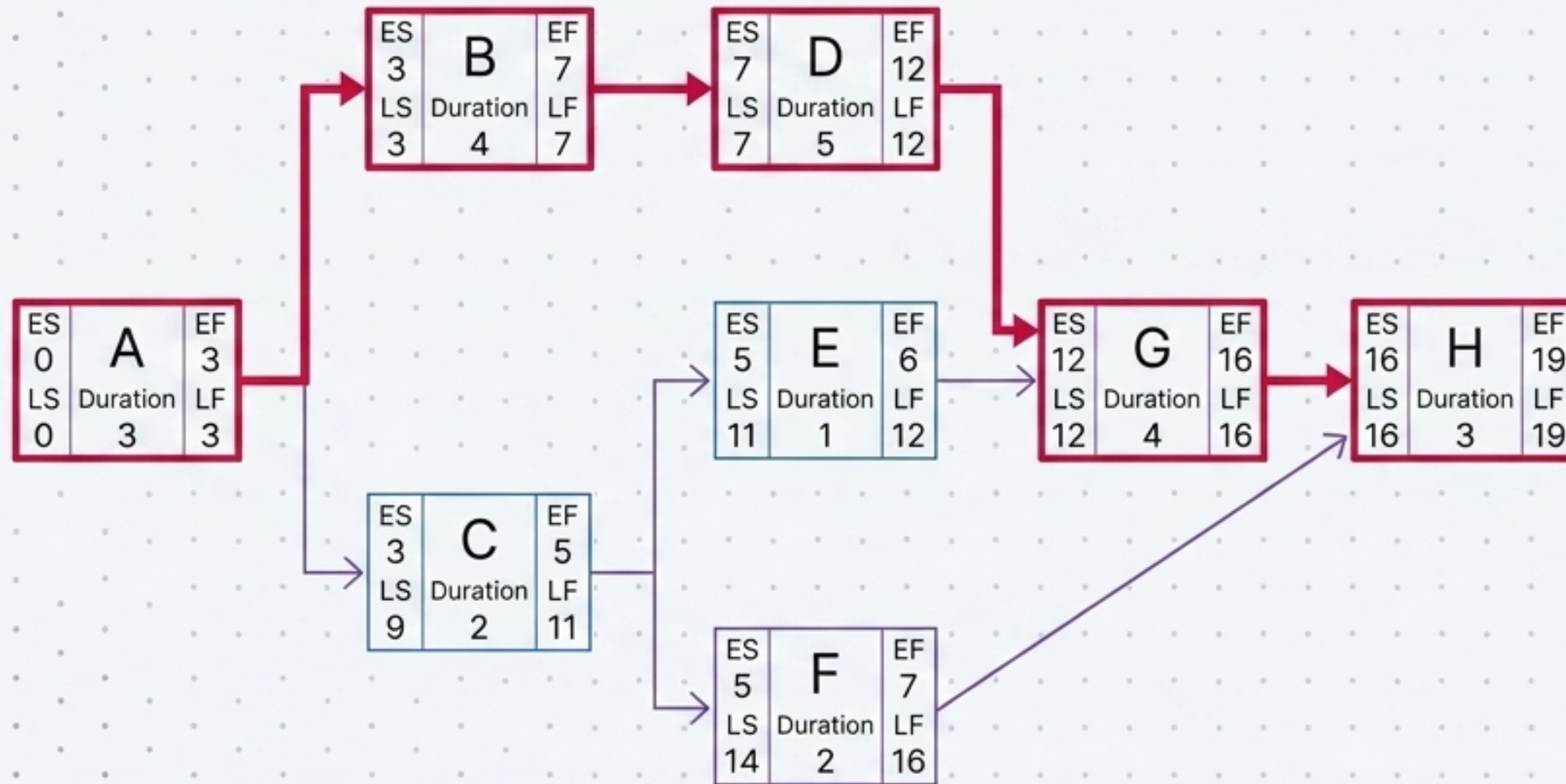
The Forward Pass: Discovering Minimum Project Time



The Backward Pass: Establishing Absolute Deadlines



Exposing the Critical Path



All Possible Paths	
Path 1:	A, B, D, G, H = 19 Days **CRITICAL**
Path 2:	A, C, E, G, H = 13 Days
Path 3:	A, C, F, H = 10 Days

The longest path through the network dictates the shortest possible time the project can be completed. This is the **Critical Path**.

The Flexibility Framework: Understanding Float

Total Float (TF)

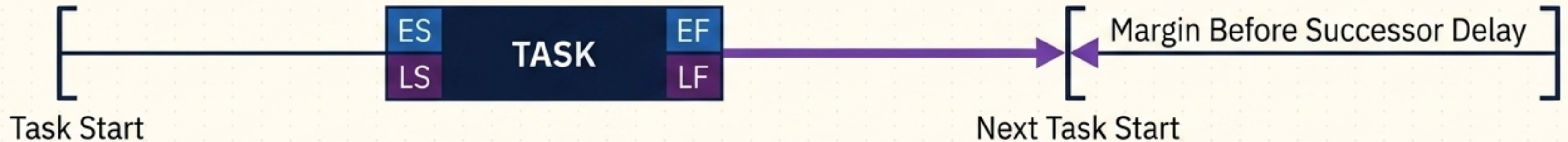
Formula: $LF - EF$ or $LS - ES$



The total margin a task can slip before delaying the entire project.

Free Float (FF)

Formula: $\text{Min}(\text{ES of Successors}) - \text{ES of Activity} - \text{Duration}$



The margin a task can slip without delaying the immediate next step in the chain.

Key Insight: Nodes on the Critical Path **ALWAYS** have **zero Total Float** and **zero Free Float**. They are **rigid**.

The Engineer's Workbench: Your Next Blueprint

Activity	Predecessor	Duration
A	-	5
B	A	4
C	A	5
D	B	6
E	C	3
F	D,E	4

The Mission

- [] Draw the Activity Diagram
- [] Execute Forward Pass (ES/EF)
- [] Execute Backward Pass (LS/LF)
- [] Identify all paths and isolate the Critical Path
- [] Calculate slack values to map project elasticity

The tools to map complexity and eliminate unpredictable delays are now in your hands.